

Top Quark Signal in the Cosmological Higgs Collider

1 Setup

We calculate the top-quark contribution to the connected Higgs four-point function in the collapsed, or s -channel squeezed, limit. As in the gauge-boson calculation, the Higgs is treated as massless during inflation. Its Schwinger–Keldysh bulk-to-boundary propagator is:

$$G_a(k; \tau) = \frac{H^2}{2k^3}(1 - iak\tau)e^{iak\tau}, \quad (1)$$

and the scale factor is $a(\tau) = -(H\tau)^{-1}$.

In unitary gauge the top Yukawa interaction is

$$\mathcal{L}_t = it^\dagger \bar{\sigma}^\mu \partial_\mu t + it^{c\dagger} \bar{\sigma}^\mu \partial_\mu t^c - a(\tau) [m_t + g_t \delta h] (tt^c + t^\dagger t^{c\dagger}), \quad (2)$$

$$g_t \equiv \frac{y_t}{\sqrt{2}}, \quad m_t = g_t h_0, \quad \tilde{\nu} \equiv \frac{m_t}{H}. \quad (3)$$

Here t and t^c are two left-handed Weyl fields and have already been conformally rescaled by $a^{3/2}$. Hence every Yukawa vertex carries one power of $a(\tau)$, rather than $a^4(\tau)$.

1.1 From the Dirac top to two Majorana spinors

The fermion formalism of Ref. [2] is written for a two-component Majorana field. It can be applied to the top without changing the physics by the unitary Takagi rotation

$$t = \frac{N_1 + iN_2}{\sqrt{2}}, \quad t^c = \frac{N_1 - iN_2}{\sqrt{2}}. \quad (4)$$

Because the contraction of two Weyl spinors is symmetric, $N_1 N_2 = N_2 N_1$, this gives

$$tt^c = \frac{1}{2}(N_1 N_1 + N_2 N_2). \quad (5)$$

Thus Eq. (2) becomes

$$\mathcal{L}_t = \sum_{A=1}^2 \left[iN_A^\dagger \bar{\sigma}^\mu \partial_\mu N_A - \frac{am_t}{2}(N_A N_A + N_A^\dagger N_A^\dagger) - \frac{ag_t}{2} \delta h (N_A N_A + N_A^\dagger N_A^\dagger) \right]. \quad (6)$$

Both N_A therefore have the same positive Majorana mass and the same Higgs coupling. This is preferable to diagonalizing the mass matrix with eigenvalues $\pm m_t$: the phase in Eq. (4) puts the result directly in the positive-mass convention used in Ref. [2].

There is no additional factor of two in a closed top loop. A single Majorana functional determinant is one half of a Dirac determinant, while the sum over $A = 1, 2$ supplies the opposite factor two. The net multiplicity is consequently the standard color factor $N_c = 3$, together with the usual minus sign for a closed fermion loop.

For momentum $\mathbf{k} = k(\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$, we use the helicity eigenspinors

$$h_+(\theta, \phi) = \begin{pmatrix} e^{-i\phi/2} \cos(\theta/2) \\ e^{i\phi/2} \sin(\theta/2) \end{pmatrix}, \quad h_-(\theta, \phi) = \begin{pmatrix} -e^{-i\phi/2} \sin(\theta/2) \\ e^{i\phi/2} \cos(\theta/2) \end{pmatrix}, \quad (7)$$

The mode functions:

$$u_+(k, \tau) = \frac{\tilde{\nu}}{\sqrt{-2k\tau}} W_{-1/2, i\tilde{\nu}}(2ik\tau), \quad u_-(k, \tau) = \frac{1}{\sqrt{-2k\tau}} W_{1/2, i\tilde{\nu}}(2ik\tau), \quad (8)$$

$$v_+(k, \tau) = \frac{1}{\sqrt{-2k\tau}} W_{1/2, i\tilde{\nu}}(2ik\tau), \quad v_-(k, \tau) = \frac{\tilde{\nu}}{\sqrt{-2k\tau}} W_{-1/2, i\tilde{\nu}}(2ik\tau). \quad (9)$$

1.2 Fermion propagators and their nonlocal parts

There are three useful two-component propagators,

$$D_{ab\alpha\beta} \equiv \langle N_\alpha N_\beta^\dagger \rangle_{ab}, \quad \widehat{D}_{ab\alpha\beta} \equiv \langle N_\alpha N_\beta \rangle_{ab}, \quad \check{D}_{ab\dot{\alpha}\dot{\beta}} \equiv \langle N_{\dot{\alpha}}^\dagger N_{\dot{\beta}}^\dagger \rangle_{ab}. \quad (10)$$

Following Ref. [2], we call D , $\widehat{D}$, and $\check{D}$ the type-1, type-2, and type-3 propagators, respectively. The nonlocal late-time parts of the symmetric type-2 and type-3 propagators are especially simple:

$$\left[\widehat{D}^{(S)\alpha}_{\beta}(k; \tau_1, \tau_2) \right]_{NL\text{oc}} = \sum_{f=\pm} \kappa_f(\tilde{\nu}) (4k^2 \tau_1 \tau_2)^{if\tilde{\nu}} \delta^\alpha_\beta, \quad (11)$$

$$\left[\check{D}^{(S)\dot{\alpha}}_{\dot{\beta}}(k; \tau_1, \tau_2) \right]_{NL\text{oc}} = \sum_{f=\pm} \kappa_f(\tilde{\nu}) (4k^2 \tau_1 \tau_2)^{if\tilde{\nu}} \delta^{\dot{\alpha}}_{\dot{\beta}}, \quad (12)$$

where

$$\kappa_f(\tilde{\nu}) = \frac{\tilde{\nu} \Gamma^2(-2if\tilde{\nu})}{\Gamma(1 - if\tilde{\nu}) \Gamma(-if\tilde{\nu})}, \quad \kappa_- = \kappa_+^*. \quad (13)$$

At zero chemical potential the helicity tensor in the unraised propagator reduces to the invariant antisymmetric tensor,

$$h_{+\alpha} h_{-\beta} - h_{-\alpha} h_{+\beta} = \epsilon_{\alpha\beta}. \quad (14)$$

It follows that the two Wightman orderings have the same type-2 nonlocal part.

The type-1 propagator behaves differently. At zero chemical potential its symmetric nonlocal part vanishes, while its antisymmetric part is proportional to $\sigma \cdot \hat{\mathbf{k}}$. Consequently a pair of type-1 soft lines is not covered by the factorized symmetric prescription of Ref. [2]. We return to this point in Sec. 4.

2 Box diagram and classification of the sixteen contractions

For one cyclic ordering of the external Higgs legs, the box has the schematic SK form

$$\langle \delta h^4 \rangle'_{\text{Box}} = -N_c (ig_t)^4 \sum_{a,b,c,d=\pm} (abcd) \int \prod_{j=1}^4 d\tau_j a(\tau_j) G_{a_j}(k_j; \tau_j) \int \frac{d^3 \mathbf{q}}{(2\pi)^3} \mathcal{S}_{abcd}, \quad (15)$$

where $(a_1, a_2, a_3, a_4) = (a, b, c, d)$ and $\mathcal{S}_{abcd}$ denotes the cyclic spinor contraction of the four fermion propagators. Crossed cyclic orderings are obtained by permuting the external labels.

The nonlocal s -channel signal arises when the lines with momenta $\mathbf{q}$ and $\mathbf{K} = \mathbf{q} + \mathbf{k}_s$ are soft. At every Yukawa vertex one may choose either NN or $N^\dagger N^\dagger$. We denote these choices by N and $\bar{N}$. They give $2^4 = 16$ Weyl contractions for a fixed cyclic ordering of the four external Higgs legs. These are terms in the Majorana decomposition of one box topology, not sixteen different momentum topologies. For compactness, on an oriented edge we write

$$NN : \widehat{D}, \quad N\bar{N} : D, \quad \bar{N}N : \bar{D}, \quad \bar{N}\bar{N} : \check{D}. \quad (16)$$

The reversed type-1 propagator is not an additional independent object:

$$\bar{D}_{ab}^{\dot{\alpha}\beta}(\mathbf{k}; \tau, \tau') \equiv -D_{ba}^{\beta\dot{\alpha}}(-\mathbf{k}; \tau', \tau). \quad (17)$$

This relation fixes the Fermi sign and the SK/time reversal for every $\bar{D}$ entry. The hard edges are (12) and (34), and the soft edges are (23) and (41). Before making either a symmetric or an antisymmetric soft-line replacement, the complete list is

No.	$(r_1 r_2 r_3 r_4)$	$(12, 23, 34, 41)$	soft class	n_1
1	$NNNN$	$(\widehat{D}, \widehat{D}, \widehat{D}, \widehat{D})$	AA	0
2	$NNN\bar{N}$	$(\widehat{D}, \widehat{D}, D, \bar{D})$	AN	1
3	$NN\bar{N}N$	$(\widehat{D}, D, \bar{D}, \widehat{D})$	NA	1
4	$NN\bar{N}\bar{N}$	$(\widehat{D}, D, \check{D}, \bar{D})$	NN	2
5	$N\bar{N}NN$	$(D, \bar{D}, \widehat{D}, \widehat{D})$	NA	1
6	$N\bar{N}N\bar{N}$	$(D, \bar{D}, D, \bar{D})$	NN	2
7	$N\bar{N}\bar{N}N$	$(D, \check{D}, \bar{D}, \widehat{D})$	AA	0
8	$N\bar{N}\bar{N}\bar{N}$	$(D, \check{D}, \check{D}, \bar{D})$	AN	1
9	$\bar{N}NNN$	$(\bar{D}, \widehat{D}, \widehat{D}, D)$	AN	1
10	$\bar{N}NN\bar{N}$	$(\bar{D}, \widehat{D}, D, \check{D})$	AA	0
11	$\bar{N}N\bar{N}N$	$(\bar{D}, D, \bar{D}, D)$	NN	2
12	$\bar{N}N\bar{N}\bar{N}$	$(\bar{D}, D, \check{D}, \check{D})$	NA	1
13	$\bar{N}\bar{N}NN$	$(\check{D}, \bar{D}, \widehat{D}, D)$	NN	2
14	$\bar{N}\bar{N}N\bar{N}$	$(\check{D}, \bar{D}, D, \check{D})$	NA	1
15	$\bar{N}\bar{N}\bar{N}N$	$(\check{D}, \check{D}, \bar{D}, D)$	AN	1
16	$\bar{N}\bar{N}\bar{N}\bar{N}$	$(\check{D}, \check{D}, \check{D}, \check{D})$	AA	0

Here A is only a compact class label for a type-2 $\widehat{D}$ or type-3 $\check{D}$ soft line, while N denotes a type-1 D or $\bar{D}$ soft line; n_1 counts the type-1 soft lines.

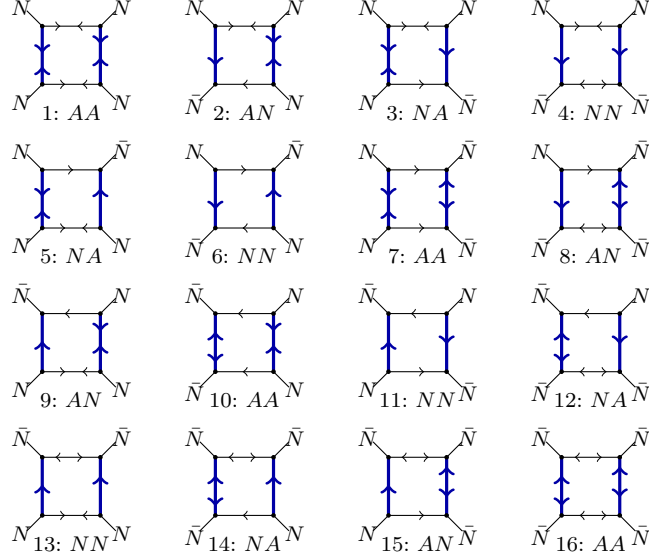


Figure 1: The sixteen Weyl contractions for one cyclic box, using the fermion-arrow convention of Ref. [2]. A type-1 line has one arrow directed from N to $N^\dagger$; the two arrows on a type-2 line point inward, while those on a type-3 line point outward. The horizontal edges (12) and (34) are hard; the blue vertical edges (23) and (41) are soft. The label below each diagram is its soft class.

3 Symmetric soft-line contribution

We first replace both soft lines by their symmetric nonlocal parts. At zero chemical potential,

$$[D^{(S)}]_{NLoc} = 0, \quad [\widehat{D}^{(S)}]_{NLoc} \neq 0, \quad [\check{D}^{(S)}]_{NLoc} \neq 0. \quad (18)$$

Consequently

$$C_\ell^{(S)} = 0, \quad \ell \in \{2, 3, 4, 5, 6, 8, 9, 11, 12, 13, 14, 15\}, \quad (19)$$

$$\mathcal{G}_{AA} = \{1, 7, 10, 16\}. \quad (20)$$

Thus twelve rows vanish and only four rows survive. This condition concerns the *soft* edges only: the hard edges of rows 1, 7, 10, and 16 are respectively (2, 2), (1, 1), (1, 1), and (3, 3).

3.1 Loop momentum integral and bubble subgraph

Following the gauge-boson calculation, we keep the two hard propagators unevaluated and perform the soft loop momentum first. For a surviving word $w_\ell = (r_1 r_2 r_3 r_4)$, define

$$\mathcal{H}_{\ell,abcd}(\tau_i) = \text{tr}_2 [\mathcal{D}_{ab}^{r_1 r_2}(\mathbf{k}_1; \tau_1, \tau_2) \Delta^{r_2 r_3} \mathcal{D}_{cd}^{r_3 r_4}(\mathbf{k}_3; \tau_3, \tau_4) \Delta^{r_4 r_1}], \quad (21)$$

where the raised-index type-2/type-3 soft matrices are $\Delta^{NN} = \Delta^{\bar{N}\bar{N}} = \sigma^0$. Before doing the $\mathbf{q}$ integral,

$$\mathcal{LOOP}_\ell^{(S)} = \sum_{f=\pm} \kappa_f^2 2^{4if\tilde{\nu}} \left[\prod_{j=1}^4 (-\tau_j)^{if\tilde{\nu}} \right] \mathcal{H}_{\ell,abcd}(\tau_i) \int \frac{d^3 \mathbf{q}}{(2\pi)^3} q^{2if\tilde{\nu}} K^{2if\tilde{\nu}}. \quad (22)$$

The loop integral therefore separates into a bubble subgraph and one extra time factor at each vertex:

$$\mathcal{LOOP}_\ell^{(S)} = \sum_{f=\pm} \left[\prod_{j=1}^4 (-\tau_j)^{if\tilde{\nu}} \right] \mathcal{B}_f^{(S)}(k_s) \mathcal{H}_{\ell,abcd}(\tau_i), \quad (23)$$

where

$$\mathcal{B}_f^{(S)}(k_s) = \kappa_f^2 2^{4if\tilde{\nu}} \int \frac{d^3\mathbf{q}}{(2\pi)^3} q^{2if\tilde{\nu}} K^{2if\tilde{\nu}} \quad (24)$$

$$= \kappa_f^2 2^{4if\tilde{\nu}} \frac{k_s^{3+4if\tilde{\nu}}}{(4\pi)^{3/2}} f_0(-if\tilde{\nu}, -if\tilde{\nu}). \quad (25)$$

In the notation of the gauge-boson calculation,

$$f_0(x, y) = \Gamma \left[\begin{matrix} x + y - \frac{3}{2}, & \frac{3}{2} - x, & \frac{3}{2} - y \\ 3 - x - y, & x, & y \end{matrix} \right]. \quad (26)$$

The factor $2^{4if\tilde{\nu}}$ follows directly from the normalization in Eq. (12).

3.2 Left and right subgraphs

The extra powers in Eq. (23) are now absorbed into the left and right time integrals. We first write these subgraphs before introducing any seed. For the chirality word $w_\ell = (r_1 r_2 r_3 r_4)$, their matrix-valued integral definitions are

$$\begin{aligned} \mathcal{T}_{f,\ell}^{(L)} &= (ig_t)^2 \sum_{a,b=\pm} ab \int_{-\infty}^0 d\tau_1 d\tau_2 a(\tau_1) a(\tau_2) G_a(k_1; \tau_1) G_b(k_2; \tau_2) \\ &\quad \times (-\tau_1)^{if\tilde{\nu}} (-\tau_2)^{if\tilde{\nu}} \mathcal{D}_{ab}^{r_1 r_2}(\mathbf{k}_1; \tau_1, \tau_2), \end{aligned} \quad (27a)$$

$$\begin{aligned} \mathcal{T}_{f,\ell}^{(R)} &= (ig_t)^2 \sum_{c,d=\pm} cd \int_{-\infty}^0 d\tau_3 d\tau_4 a(\tau_3) a(\tau_4) G_c(k_3; \tau_3) G_d(k_4; \tau_4) \\ &\quad \times (-\tau_3)^{if\tilde{\nu}} (-\tau_4)^{if\tilde{\nu}} \mathcal{D}_{cd}^{r_3 r_4}(\mathbf{k}_3; \tau_3, \tau_4). \end{aligned} \quad (27b)$$

The SK weights ab and cd are included here, whereas the soft coefficient κ_f^2 belongs to the bubble subgraph. Inserting the scale factor and the two massless Higgs wavefunctions into the first line gives

$$\begin{aligned} \mathcal{T}_{f,\ell}^{(L)} &= (ig_t)^2 \frac{H^2}{4k_1^3 k_2^3} \sum_{a,b=\pm} ab \int_{-\infty}^0 d\tau_1 d\tau_2 (-\tau_1)^{p_f} (-\tau_2)^{p_f} \\ &\quad \times (1 - iak_1\tau_1)(1 - ibk_2\tau_2) e^{iak_1\tau_1 + ibk_2\tau_2} \mathcal{D}_{ab}^{r_1 r_2}(\mathbf{k}_1; \tau_1, \tau_2), \quad p_f = -1 + if\tilde{\nu}. \end{aligned} \quad (28)$$

The right subgraph is obtained by $(1, 2, a, b, r_1, r_2) \rightarrow (3, 4, c, d, r_3, r_4)$.

Now define the two hard scales

$$k_L \equiv \frac{k_{12}}{2}, \quad k_R \equiv \frac{k_{34}}{2}. \quad (29)$$

At leading order in the collapsed limit, $k_1 = k_2 = k_L + \mathcal{O}((\cdot)k_s)$ and $k_3 = k_4 = k_R + \mathcal{O}((\cdot)k_s)$. With $z_1 = -k_L\tau_1$ and $z_2 = -k_L\tau_2$, Eq. (28) becomes

$$\begin{aligned} \mathcal{T}_{f,\ell}^{(L)} &= (ig_t)^2 \frac{H^2}{4k_1^3 k_2^3} k_L^{-2if\tilde{\nu}} \sum_{a,b=\pm} ab \int_0^\infty dz_1 dz_2 z_1^{p_f} z_2^{p_f} (1 + ia z_1)(1 + ib z_2) \\ &\quad \times e^{-ia z_1 - ib z_2} \mathcal{D}_{ab}^{r_1 r_2}(\hat{\mathbf{k}}_1; z_1, z_2) + \mathcal{O}((\cdot)k_s/k_i). \end{aligned} \quad (30)$$

The dimensionless propagator on the last line is the one used in Ref. [2]; conformal rescaling makes its momentum dependence a function of $k_L\tau_{1,2} = -z_{1,2}$ and the direction $\hat{\mathbf{k}}_1$. Retaining unequal k_1/k_L and k_2/k_L would merely give energy-deformed seeds and contributes beyond the leading collapsed result considered here.

We can now read off the You seeds. For endpoint chiralities $r, s \in \{N, \bar{N}\}$ define

$$\mathcal{I}_{ab}^{rs;p_1 p_2}(\hat{\mathbf{k}}) = \int_0^\infty dz_1 dz_2 e^{-ia z_1 - ib z_2} z_1^{p_1} z_2^{p_2} \mathcal{D}_{ab}^{rs}(\hat{\mathbf{k}}; z_1, z_2). \quad (31)$$

Here $\mathcal{D}^{NN} = \widehat{D}$, $\mathcal{D}^{N\bar{N}} = D$, $\mathcal{D}^{\bar{N}N} = \bar{D}$, and $\mathcal{D}^{\bar{N}\bar{N}} = \check{D}$. Local end-point terms are defined by analytic continuation and do not affect the nonanalytic signal. Expanding the two Higgs polynomials in Eq. (30) gives exactly

$$\mathcal{H}_{f;ab}^{rs} = \mathcal{I}_{ab}^{rs;p_f p_f} + ia \mathcal{I}_{ab}^{rs;p_f+1, p_f} + ib \mathcal{I}_{ab}^{rs;p_f, p_f+1} - ab \mathcal{I}_{ab}^{rs;p_f+1, p_f+1}, \quad (32)$$

$$\mathcal{Q}_f^{rs}(\hat{\mathbf{k}}) = \sum_{a,b=\pm} ab \mathcal{H}_{f;ab}^{rs}(\hat{\mathbf{k}}). \quad (33)$$

Therefore the two original time integrals reduce to

$$\mathcal{T}_{f,\ell}^{(L)} = (ig_t)^2 \frac{H^2}{4k_1^3 k_2^3} \left(\frac{k_{12}}{2} \right)^{-2if\tilde{\nu}} \mathcal{Q}_f^{r_1 r_2}(\hat{\mathbf{k}}_1), \quad (34)$$

$$\mathcal{T}_{f,\ell}^{(R)} = (ig_t)^2 \frac{H^2}{4k_3^3 k_4^3} \left(\frac{k_{34}}{2} \right)^{-2if\tilde{\nu}} \mathcal{Q}_f^{r_3 r_4}(\hat{\mathbf{k}}_3). \quad (35)$$

Appendix A gives the type-1, type-2, and type-3 seed matrices required by Eq. (33). Thus no hard time integral remains. For the type-1 hard edges of rows 7 and 10 we abbreviate

$$\mathcal{Q}_f^{N\bar{N}}(\hat{\mathbf{k}}) = Q_{f,0}\sigma^0 + Q_{f,1}\hat{\mathbf{k}} \cdot \boldsymbol{\sigma}. \quad (36)$$

3.3 Factorized form before the spinor contraction

Let $w_\ell = (r_1 r_2 r_3 r_4)$ denote the chirality word in row ℓ . After the bubble integral has been done, the symmetric contribution takes exactly the same left–bubble–right form as the gauge-boson box,

$$\langle \delta h^4 \rangle'_{\text{Box}} \Big|_S = -N_c \sum_{\ell \in \mathcal{G}_{AA}} \sum_{f=\pm} \mathcal{T}_{f,\ell}^{(L)} \mathcal{B}_f^{(S)} \mathcal{T}_{f,\ell}^{(R)} \Big|_{\text{spinor contraction}}. \quad (37)$$

The scalar bubble coefficient is Eq. (25); all spinor indices reside in the two hard blocks. Since a symmetric type-2 or type-3 soft propagator becomes a Kronecker delta after raising the index contracted at the adjacent Yukawa vertex, no direction associated with $\mathbf{q}$ or $\mathbf{K}$ remains in this sector. The spinor contraction can therefore be completed before doing either hard time integral.

3.4 Four spinor contractions and the closed symmetric result

We now evaluate rows 1, 7, 10, 16 separately. This also fixes the relative sign between the type-1 and type-2/type-3 hard lines. Let

$$E_{\alpha\beta} = \epsilon_{\alpha\beta}, \quad \widehat{\mathbf{Q}}_f = \widehat{\mathbf{Q}}_f E^{-1}, \quad \check{\mathbf{Q}}_f = E^{-1} \check{\mathbf{Q}}_f, \quad (38)$$

where the placement of E^{-1} is fixed by whether the indices are undotted or dotted. These are just the mixed-index matrices obtained by contracting the hard propagator with the two Yukawa tensors. In the $\mathbf{k} \parallel \hat{\mathbf{z}}$ frame define their four pairs of radial coefficients by

$$\begin{aligned} \widehat{\mathbf{Q}}_+(\hat{\mathbf{z}}) &= \hat{q}_0 \sigma^0 + \hat{q}_1 \sigma^3, & \check{\mathbf{Q}}_+(\hat{\mathbf{z}}) &= \check{q}_0 \sigma^0 + \check{q}_1 \bar{\sigma}^3, \\ \mathbf{Q}_+(\hat{\mathbf{z}}) &= q_0 \sigma^0 + q_1 \sigma^3, & \bar{\mathbf{Q}}_+(\hat{\mathbf{z}}) &= \bar{q}_0 \sigma^0 + \bar{q}_1 \bar{\sigma}^3, \quad \bar{\sigma}^i = -\sigma^i. \end{aligned} \quad (39)$$

Here $\mathbf{Q}$ and $\bar{\mathbf{Q}}$ are respectively the D and $\bar{D}$ hard blocks. The eight quantities in Eq. (39) depend only on $\tilde{\nu}$ at the leading collapsed order. The rotation rule of Ref. [2] gives

$$\sigma^3 \longrightarrow \hat{\mathbf{k}} \cdot \boldsymbol{\sigma}, \quad \bar{\sigma}^3 \longrightarrow \hat{\mathbf{k}} \cdot \bar{\boldsymbol{\sigma}}. \quad (40)$$

We choose the same angles as in the gauge-boson box,

$$\begin{aligned} \mathbf{k}_s &= k_s(0, 0, 1), \\ \mathbf{k}_1 &= k_1(\sin \theta_1, 0, \cos \theta_1), \\ \mathbf{k}_3 &= k_3(\sin \theta_3 \cos \phi_3, \sin \theta_3 \sin \phi_3, \cos \theta_3), \end{aligned} \quad (41)$$

and use the single invariant

$$c_{13} \equiv \hat{\mathbf{k}}_1 \cdot \hat{\mathbf{k}}_3 = \cos \theta_1 \cos \theta_3 + \sin \theta_1 \sin \theta_3 \cos \phi_3. \quad (42)$$

The two raised soft lines are identities. Using $\text{tr}(\sigma^i \sigma^j) = 2\delta^{ij}$ and $\text{tr}(\sigma^i \bar{\sigma}^j) = -2\delta^{ij}$, the four contractions are

$$\mathcal{W}_{+,1} = \text{tr}_2 \left[\widehat{\mathbf{Q}}_+(\hat{\mathbf{k}}_1) \widehat{\mathbf{Q}}_+(\hat{\mathbf{k}}_3) \right] = 2 \left(\hat{q}_0^2 + \hat{q}_1^2 c_{13} \right), \quad (43a)$$

$$\mathcal{W}_{+,7} = \text{tr}_2 \left[\mathbf{Q}_+(\hat{\mathbf{k}}_1) \bar{\mathbf{Q}}_+(\hat{\mathbf{k}}_3) \right] = 2 \left(q_0 \bar{q}_0 - q_1 \bar{q}_1 c_{13} \right), \quad (43b)$$

$$\mathcal{W}_{+,10} = \text{tr}_2 \left[\bar{\mathbf{Q}}_+(\hat{\mathbf{k}}_1) \mathbf{Q}_+(\hat{\mathbf{k}}_3) \right] = 2 \left(\bar{q}_0 q_0 - \bar{q}_1 q_1 c_{13} \right), \quad (43c)$$

$$\mathcal{W}_{+,16} = \text{tr}_2 \left[\check{\mathbf{Q}}_+(\hat{\mathbf{k}}_1) \check{\mathbf{Q}}_+(\hat{\mathbf{k}}_3) \right] = 2 \left(\check{q}_0^2 + \check{q}_1^2 c_{13} \right). \quad (43d)$$

Thus rows 7 and 10 are equal after the radial coefficients have been extracted, but both must be retained: they are the two opposite fermion-flow assignments. Equations (43) also show directly why the symmetric angular dependence cannot contain $\cos 2\phi_3$, $\cos \theta_1$, or $\cos \theta_3$ separately.

For completeness, we now write the hard quantities in terms of the integral seeds of Ref. [2]. With $p = -1 + i\tilde{\nu}$, define for $X = D, \bar{D}, \widehat{D}, \check{D}$

$$\mathbf{Q}_+^X(\hat{\mathbf{k}}) = \sum_{a,b=\pm} ab \left[\mathbf{I}_{ab}^{X;p,p} + ia \mathbf{I}_{ab}^{X;p+1,p} + ib \mathbf{I}_{ab}^{X;p,p+1} - ab \mathbf{I}_{ab}^{X;p+1,p+1} \right] (\hat{\mathbf{k}}). \quad (44)$$

This is the scalar-Yukawa analogue of the subdiagram construction in Ref. [2]: the four terms $(m, n) = (0, 0), (1, 0), (0, 1), (1, 1)$ come from the two massless Higgs polynomials. For $X = \widehat{D}, \check{D}$, the index-raising operation in Eq. (38) is understood on the right-hand side. More explicitly, at leading collapsed order we may set $k_1 = k_2 = k_{12}/2 + \mathcal{O}(\epsilon k_s)$ and $k_3 = k_4 = k_{34}/2 + \mathcal{O}(\epsilon k_s)$. Putting

$$\mathcal{N}_L = (ig_t)^2 \frac{H^2}{4k_1^3 k_2^3} \left(\frac{k_{12}}{2} \right)^{-2i\widetilde{\nu}}, \quad \mathcal{N}_R = (ig_t)^2 \frac{H^2}{4k_3^3 k_4^3} \left(\frac{k_{34}}{2} \right)^{-2i\widetilde{\nu}}, \quad (45)$$

the four pairs of left and right subdiagrams are

$$\begin{aligned} (\mathcal{T}_{+,1}^{(L)}, \mathcal{T}_{+,1}^{(R)}) &= (\mathcal{N}_L \widehat{\mathbf{Q}}_+(\widehat{\mathbf{k}}_1), \mathcal{N}_R \widehat{\mathbf{Q}}_+(\widehat{\mathbf{k}}_3)), \\ (\mathcal{T}_{+,7}^{(L)}, \mathcal{T}_{+,7}^{(R)}) &= (\mathcal{N}_L \mathbf{Q}_+(\widehat{\mathbf{k}}_1), \mathcal{N}_R \overline{\mathbf{Q}}_+(\widehat{\mathbf{k}}_3)), \\ (\mathcal{T}_{+,10}^{(L)}, \mathcal{T}_{+,10}^{(R)}) &= (\mathcal{N}_L \overline{\mathbf{Q}}_+(\widehat{\mathbf{k}}_1), \mathcal{N}_R \mathbf{Q}_+(\widehat{\mathbf{k}}_3)), \\ (\mathcal{T}_{+,16}^{(L)}, \mathcal{T}_{+,16}^{(R)}) &= (\mathcal{N}_L \check{\mathbf{Q}}_+(\widehat{\mathbf{k}}_1), \mathcal{N}_R \check{\mathbf{Q}}_+(\widehat{\mathbf{k}}_3)). \end{aligned} \quad (46)$$

Thus the subdiagrams in Eqs. (43) are now finite sums of the You integral seeds, rather than unevaluated time integrals. The type-1 and type-2 matrices $\mathbf{I}^D$ and $\mathbf{I}^{\widehat{D}}$ are displayed explicitly in Eqs. (104) and (115). The other two contain no new integral,

$$\begin{aligned} \mathbf{I}_{ab}^{\widehat{D}; p_1 p_2}(\widehat{\mathbf{k}}) &= - \left[\mathbf{I}_{ba}^{D; p_2 p_1}(-\widehat{\mathbf{k}}) \right]^T, \\ \mathbf{I}_{ab}^{\check{D}; p_1 p_2}(\widehat{\mathbf{k}}) &= - \left[\mathbf{I}_{-a, -b}^{\widehat{D}; p_1^* p_2^*}(\widehat{\mathbf{k}}) \right]^*. \end{aligned} \quad (47)$$

After the index raising in Eq. (38), the radial coefficients are obtained without a further integral:

$$\begin{aligned} q_{X0} &= \frac{1}{2} \left[(\mathbf{Q}_+^X)_{11} + (\mathbf{Q}_+^X)_{22} \right], \\ q_{X1} &= \frac{\eta_X}{2} \left[(\mathbf{Q}_+^X)_{11} - (\mathbf{Q}_+^X)_{22} \right], \quad \eta_D = \eta_{\widehat{D}} = 1, \quad \eta_{\bar{D}} = \eta_{\check{D}} = -1. \end{aligned} \quad (48)$$

In Eq. (48), $q_{Di} = q_i$, $q_{\bar{D}i} = \bar{q}_i$, $q_{\widehat{D}i} = \widehat{q}_i$, and $q_{\check{D}i} = \check{q}_i$.

To make the special-function dependence explicit, all entries needed in Eqs. (44)–(48) reduce to

$$\begin{aligned}
\mathfrak{K}^{p_1 p_2}(x, y) &= e^{\frac{i\pi}{2}(p_1 - p_2)} \frac{\prod_{j=1}^2 [\Gamma(1 + p_j - i\tilde{\nu})\Gamma(1 + p_j + i\tilde{\nu})]}{2^{2+p_1+p_2}\Gamma(p_1 + x)\Gamma(p_2 + y)}, \\
\mathfrak{J}^{p_1 p_2}(x, y, z) &= i 2^{-2-p_1-p_2} e^{-\frac{i\pi}{2}(p_1+p_2)} \pi \operatorname{csch}(2\pi\tilde{\nu}) \Gamma(2 + p_1 + p_2) \\
&\quad \times \left[\frac{e^{-\pi\tilde{\nu}} \Gamma(1 + p_1 - i\tilde{\nu}) \Gamma(2 + p_1 + p_2 - 2i\tilde{\nu})}{\Gamma(x)} \right. \\
&\quad \left. \times {}_4\tilde{F}_3 \left(\begin{matrix} 2 + p_1 + p_2, 1 + p_1 - i\tilde{\nu}, y, 2 + p_1 + p_2 - 2i\tilde{\nu} \\ 2 + p_1 - i\tilde{\nu}, 3 + p_1 + p_2 - z, 1 - 2i\tilde{\nu} \end{matrix}; -1 \right) - (\tilde{\nu} \rightarrow -\tilde{\nu}) \right], \\
\mathfrak{J}^{p_1 p_2}(x, y, z) &= i 2^{-2-p_1-p_2} e^{\frac{i\pi}{2}(p_1+p_2)} \pi \operatorname{csch}(2\pi\tilde{\nu}) \Gamma(2 + p_1 + p_2) \\
&\quad \times \left[\frac{e^{\pi\tilde{\nu}} \Gamma(1 + p_1 - i\tilde{\nu}) \Gamma(2 + p_1 + p_2 - 2i\tilde{\nu})}{\Gamma(x)} \right. \\
&\quad \left. \times {}_4\tilde{F}_3 \left(\begin{matrix} 2 + p_1 + p_2, 1 + p_1 - i\tilde{\nu}, y, 2 + p_1 + p_2 - 2i\tilde{\nu} \\ 2 + p_1 - i\tilde{\nu}, 3 + p_1 + p_2 - z, 1 - 2i\tilde{\nu} \end{matrix}; -1 \right) - (\tilde{\nu} \rightarrow -\tilde{\nu}) \right].
\end{aligned} \tag{49}$$

$$\tag{51}$$

Here ${}_4\tilde{F}_3 = {}_4F_3 / \prod_{j=1}^3 \Gamma(b_j)$ is the regularized hypergeometric function. The finite list of arguments entering the type-1 block is

$$\begin{aligned}
A_1 &= \tilde{\nu}^2 \mathfrak{J}(1 + i\tilde{\nu}, 1 - i\tilde{\nu}, i\tilde{\nu}), & A_2 &= \mathfrak{J}(i\tilde{\nu}, -i\tilde{\nu}, 1 + i\tilde{\nu}), \\
B_1 &= \mathfrak{J}^{p_2 p_1}(i\tilde{\nu}, -i\tilde{\nu}, 1 + i\tilde{\nu}), & B_2 &= \tilde{\nu}^2 \mathfrak{J}^{p_2 p_1}(1 + i\tilde{\nu}, 1 - i\tilde{\nu}, i\tilde{\nu}), \\
U_1 &= \tilde{\nu}^2 \mathfrak{K}(2, 2), & U_2 &= \mathfrak{K}(1, 1), \\
V_1 &= e^{-i\pi(p_1 - p_2)} \mathfrak{K}(1, 1), & V_2 &= \tilde{\nu}^2 e^{-i\pi(p_1 - p_2)} \mathfrak{K}(2, 2),
\end{aligned} \tag{52}$$

where the superscript $p_1 p_2$ is understood on every $\mathfrak{J}$ and $\mathfrak{K}$ unless it is shown explicitly. For the type-2 block the corresponding list is

$$\begin{aligned}
S_1 &= \tilde{\nu} \mathfrak{J}(1 + i\tilde{\nu}, 1 - i\tilde{\nu}, 1 + i\tilde{\nu}), & S_2 &= \tilde{\nu} \mathfrak{J}(i\tilde{\nu}, -i\tilde{\nu}, i\tilde{\nu}), \\
S_3 &= \tilde{\nu} \mathfrak{J}^{p_2 p_1}(i\tilde{\nu}, -i\tilde{\nu}, i\tilde{\nu}), & S_4 &= \tilde{\nu} \mathfrak{J}^{p_2 p_1}(1 + i\tilde{\nu}, 1 - i\tilde{\nu}, 1 + i\tilde{\nu}), \\
W_1 &= \tilde{\nu} \mathfrak{K}(2, 1), & W_2 &= \tilde{\nu} \mathfrak{K}(1, 2).
\end{aligned} \tag{53}$$

Equations (104) and (115) specify where these entries occur in each SK matrix. Hence Eqs. (49)–(53) expose the complete mass dependence; no hard time integral is left unevaluated in the symmetric result.

Summing Eqs. (43), it is useful to define

$$\begin{aligned}
\mathcal{R}_0(\tilde{\nu}) &= \hat{q}_0^2 + \check{q}_0^2 + 2q_0 \bar{q}_0, \\
\mathcal{R}_1(\tilde{\nu}) &= \hat{q}_1^2 + \check{q}_1^2 - 2q_1 \bar{q}_1.
\end{aligned} \tag{54}$$

The remaining bubble functions can also be written entirely in Gamma functions,

$$\begin{aligned}\kappa_+ &= i \left[\frac{\Gamma(-2i\tilde{\nu})}{\Gamma(-i\tilde{\nu})} \right]^2, \\ f_0(-i\tilde{\nu}, -i\tilde{\nu}) &= \frac{\Gamma(-\frac{3}{2} - 2i\tilde{\nu})\Gamma^2(\frac{3}{2} + i\tilde{\nu})}{\Gamma(3 + 2i\tilde{\nu})\Gamma^2(-i\tilde{\nu})}.\end{aligned}\tag{55}$$

The complete symmetric amplitude for one cyclic ordering is therefore

$$C_{(S)}^{\text{Box}}(\tilde{\nu}, \theta_1, \theta_3, \phi_3) = A_0^{(S)}(\tilde{\nu}) + A_{13}^{(S)}(\tilde{\nu})c_{13},\tag{56}$$

$$\begin{aligned}A_0^{(S)}(\tilde{\nu}) &= \frac{2^{1+12i\tilde{\nu}}\kappa_+^2}{(4\pi)^{3/2}} f_0(-i\tilde{\nu}, -i\tilde{\nu})\mathcal{R}_0(\tilde{\nu}), \\ A_{13}^{(S)}(\tilde{\nu}) &= \frac{2^{1+12i\tilde{\nu}}\kappa_+^2}{(4\pi)^{3/2}} f_0(-i\tilde{\nu}, -i\tilde{\nu})\mathcal{R}_1(\tilde{\nu}).\end{aligned}\tag{57}$$

Using the gauge-boson-box normalization, the corresponding four-point function is

$$\langle \delta h^4 \rangle'_{\text{Box}}|_S = -N_c g_t^4 \frac{H^4 k_s^3}{16k_1^3 k_2^3 k_3^3 k_4^3} \left(\frac{k_s^2}{4k_{12}k_{34}} \right)^{2i\tilde{\nu}} [A_0^{(S)} + A_{13}^{(S)}c_{13}] + \text{c.c.} + \mathcal{O}\left(\frac{k_s}{k_i}\right).\tag{58}$$

This is the fermionic counterpart of the gauge-boson box result: a mass-dependent amplitude multiplying a complex power of the collapsed momentum ratio. In contrast to the vector case, the soft spinor deltas leave only the monopole and the term linear in c_{13} for a fixed cyclic ordering. The phase $2^{12i\tilde{\nu}}$ combines the normalization of the two soft propagators, the choice of hard scales $k_{12}/2$ and $k_{34}/2$, and the factor 4 in the clock argument; it is convention dependent, whereas the full expression including its complex conjugate is not.

4 Antisymmetric soft-line contribution

We next retain the antisymmetric part that was left for future work in Ref. [2]. At zero chemical potential, Eq. (2.39) and Appendix C.3 of that reference give

$$\rho_f(\tilde{\nu}) \equiv \left[\frac{\Gamma(-2if\tilde{\nu})}{\Gamma(-if\tilde{\nu})} \right]^2, \quad \kappa_f = if\rho_f,\tag{59}$$

$$[D_{ab}(k; \tau, \tau')]_{\text{NLoc}, f} = -\rho_f(4k^2\tau\tau')^{if\tilde{\nu}}\Xi_{ab}(\tau, \tau')\hat{\mathbf{k}} \cdot \sigma,\tag{60}$$

$$\Xi_{ab}(\tau, \tau') = \frac{a+b}{2} \text{sgn}(\tau - \tau') + \frac{b-a}{2}, \quad a, b = \pm 1.\tag{61}$$

Thus $\Xi_{-+} = 1$, $\Xi_{+-} = -1$, $\Xi_{++} = \text{sgn}(\tau - \tau')$, and $\Xi_{--} = -\text{sgn}(\tau - \tau')$. For the reversed type-1 line it is convenient to keep the same coefficient $-\rho_f$ and use $\hat{\mathbf{k}} \cdot \bar{\sigma} = -\hat{\mathbf{k}} \cdot \sigma$. The other two propagators obey

$$[\widehat{D}^{(A)}]_{\text{NLoc}} = 0, \quad [\check{D}^{(A)}]_{\text{NLoc}} = 0 \quad (\tilde{\lambda} = 0).\tag{62}$$

Consequently the antisymmetric contribution of rows 1, 7, 10, 16 is identically zero. The remaining twelve rows contain one or two type-1 soft lines and must be retained.

4.1 Loop momentum integral and tensor bubble subgraphs

We follow the gauge-boson box calculation and first do the soft loop momentum. Put $\mathbf{K} = \mathbf{q} + \mathbf{k}_s$ and $x = -if\tilde{\nu}$. Before this integral, the universal nonanalytic factor is

$$\mathcal{LOOP}_{f,\ell}^{soft} = \left[\prod_{j=1}^4 (-\tau_j)^{if\tilde{\nu}} \right] 2^{4if\tilde{\nu}} \int \frac{d^3\mathbf{q}}{(2\pi)^3} q^{-2x} K^{-2x} \mathcal{R}_{23} \otimes \mathcal{R}_{41}, \quad (63)$$

where $\mathcal{R} = \sigma^0$ for a type-2/type-3 line and is $\hat{\mathbf{q}} \cdot \sigma$ or $\hat{\mathbf{K}} \cdot \sigma$ for a type-1 line. A reversed type-1 line carries $\bar{\sigma} = -\sigma$; this sign will be denoted by $\zeta = -1$. Only scalar, vector, and rank-two tensor moments occur. Using the same loop seeds as the gauge-boson calculation, define

$$f_c(a, b) = \frac{1}{2}f_0\left(a + \frac{1}{2}, b - 1\right) - \frac{1}{2}f_0\left(a - \frac{1}{2}, b\right) - \frac{1}{2}f_0\left(a + \frac{1}{2}, b\right), \quad (64)$$

$$\begin{aligned} f_{cc}(a, b) &= \frac{1}{4}f_0(a + 1, b - 2) + \frac{1}{4}f_0(a - 1, b) + \frac{1}{4}f_0(a + 1, b) \\ &\quad - \frac{1}{2}f_0(a, b - 1) - \frac{1}{2}f_0(a + 1, b - 1) + \frac{1}{2}f_0(a, b). \end{aligned} \quad (65)$$

and

$$\mathcal{V}_q(x) = f_c(x, x), \quad \mathcal{V}_K(x) = -f_c(x, x), \quad (66)$$

$$\mathcal{J}_0(x) = f_0\left(x - \frac{1}{2}, x + \frac{1}{2}\right) - \frac{1}{2}f_0\left(x + \frac{1}{2}, x + \frac{1}{2}\right), \quad (67)$$

$$\mathcal{J}_L(x) = f_{cc}\left(x - \frac{1}{2}, x + \frac{1}{2}\right) + f_c\left(x, x + \frac{1}{2}\right), \quad \mathcal{J}_T(x) = \frac{\mathcal{J}_0(x) - \mathcal{J}_L(x)}{2}. \quad (68)$$

After extracting $k_s^{3+4if\tilde{\nu}}/(4\pi)^{3/2}$, the required moments are

$$\begin{aligned} \int \hat{q}_i &= \mathcal{V}_q \hat{k}_{si}, & \int \hat{K}_i &= \mathcal{V}_K \hat{k}_{si}, \\ \int \hat{K}_i \hat{q}_j &= \mathcal{J}_T(\delta_{ij} - \hat{k}_{si} \hat{k}_{sj}) + \mathcal{J}_L \hat{k}_{si} \hat{k}_{sj}. \end{aligned} \quad (69)$$

The integral signs in Eq. (69) include the common weights $q^{-2x}K^{-2x}$. The tensor trace gives the useful check $2\mathcal{J}_T + \mathcal{J}_L = \mathcal{J}_0$. Restoring the factors stripped above,

$$\mathcal{LOOP}_{f,\ell}^{soft} = \left[\prod_{j=1}^4 (-\tau_j)^{if\tilde{\nu}} \right] \frac{2^{4if\tilde{\nu}} k_s^{3+4if\tilde{\nu}}}{(4\pi)^{3/2}} \mathfrak{B}_{f,\ell} \times (\text{SK/time kernels}), \quad (70)$$

where $\mathfrak{B}_{f,\ell}$ is the vector or tensor in Eq. (69). Thus the soft loop momentum is completely integrated before treating its time-ordering kernel.

4.2 The sixteen rows in the antisymmetric sector

The following table supplies the information needed for every row. The pair in the third column is (X_L, X_R) for the hard edges (12) and (34). A symbol $D_A(K)$ or $\bar{D}_A(q)$ denotes a type-1 antisymmetric soft line and fixes both its momentum and its sign ζ , with $\zeta_D = +1$ and $\zeta_{\bar{D}} = -1$.

ℓ	word	(X_L, X_R)	edge (23)	edge (41)
1	$NNNN$	$(\hat{D}, \hat{D})$	$\hat{D}_S$	$\hat{D}_S$
2	$NNN\bar{N}$	$(\hat{D}, D)$	$\hat{D}_S$	$\bar{D}_A(q)$
3	$NN\bar{N}N$	$(\hat{D}, \bar{D})$	$D_A(K)$	$\hat{D}_S$
4	$NN\bar{N}\bar{N}$	$(\hat{D}, \check{D})$	$D_A(K)$	$\bar{D}_A(q)$
5	$N\bar{N}NN$	$(D, \hat{D})$	$\bar{D}_A(K)$	$\hat{D}_S$
6	$N\bar{N}N\bar{N}$	(D, D)	$\bar{D}_A(K)$	$\bar{D}_A(q)$
7	$N\bar{N}\bar{N}N$	$(D, \bar{D})$	$\check{D}_S$	$\hat{D}_S$
8	$N\bar{N}\bar{N}\bar{N}$	$(D, \check{D})$	$\check{D}_S$	$\bar{D}_A(q)$
9	$\bar{N}NNN$	$(\bar{D}, \hat{D})$	$\hat{D}_S$	$D_A(q)$
10	$\bar{N}NN\bar{N}$	$(\bar{D}, D)$	$\hat{D}_S$	$\check{D}_S$
11	$\bar{N}N\bar{N}N$	$(\bar{D}, \bar{D})$	$D_A(K)$	$D_A(q)$
12	$\bar{N}N\bar{N}\bar{N}$	$(\bar{D}, \check{D})$	$D_A(K)$	$\check{D}_S$
13	$\bar{N}\bar{N}NN$	$(\check{D}, \hat{D})$	$\bar{D}_A(K)$	$D_A(q)$
14	$\bar{N}\bar{N}N\bar{N}$	$(\check{D}, D)$	$\bar{D}_A(K)$	$\check{D}_S$
15	$\bar{N}\bar{N}\bar{N}N$	$(\check{D}, \bar{D})$	$\check{D}_S$	$D_A(q)$
16	$\bar{N}\bar{N}\bar{N}\bar{N}$	$(\check{D}, \check{D})$	$\check{D}_S$	$\check{D}_S$

Rows 1, 7, 10, 16 contain no antisymmetric line and hence contribute zero to the present sector. The remaining rows fall into

$$\mathcal{G}_{AN} = \{2, 8, 9, 15\}, \quad \mathcal{G}_{NA} = \{3, 5, 12, 14\}, \quad \mathcal{G}_{NN} = \{4, 6, 11, 13\}, \quad (71)$$

where AN has a type-1 line on (41), NA has one on (23), and NN has one on each soft edge. This accounts for all $4 + 4 + 4 + 4 = 16$ rows.

4.3 Time kernels and energy-deformed You seeds

The sign kernel is separated by the exact spectral identity

$$\text{sgn}(\tau - \tau') = \frac{1}{i\pi} \text{PV} \int_{-\infty}^{+\infty} \frac{d\omega}{\omega} e^{i\omega(\tau - \tau')}. \quad (72)$$

For each fixed pair of SK labels introduce

$$C_{uv}^{(0)} = \frac{v - u}{2}, \quad C_{uv}^{(1)} = \frac{u + v}{2}, \quad \Xi_{uv} = C_{uv}^{(0)} + C_{uv}^{(1)} \text{sgn}(\tau - \tau'). \quad (73)$$

Thus an opposite-SK type-1 line selects the algebraic term $C^{(0)}$, while an equal-SK line selects one Hilbert transform through $C^{(1)}$. This observation avoids any double counting of SK sums.

The hard blocks must therefore be kept branch resolved. Define the energy-deformed Yukawa seed, without an SK sum, by

$$\mathbf{Q}_{f;ab}^X(\hat{\mathbf{k}}; u, v) \equiv \int_0^\infty dz_1 dz_2 z_1^{p_f} z_2^{p_f} (1 + ia z_1)(1 + ib z_2) e^{-iaz_1 - ibz_2 + iuz_1 + ivz_2} \mathbf{D}_{ab}^X(\hat{\mathbf{k}}; z_1, z_2), \quad (74)$$

where $p_f = -1 + if\tilde{\nu}$. It is important that Eq. (74) contains no factor ab ; the four branch labels are summed only after the cross-edge kernels have been inserted. The dependence on the special

functions of Ref. [2] is made explicit by the convergent, $i\epsilon$ -regulated expansion

$$\mathbf{Q}_{f;ab}^X(u, v) = \sum_{r,s=0}^{\infty} \frac{(iu)^r (iv)^s}{r!s!} \left[\mathbf{I}_{ab}^{X;p_f+r,p_f+s} + ia \mathbf{I}_{ab}^{X;p_f+r+1,p_f+s} + ib \mathbf{I}_{ab}^{X;p_f+r,p_f+s+1} - ab \mathbf{I}_{ab}^{X;p_f+r+1,p_f+s+1} \right]. \quad (75)$$

Every matrix $\mathbf{I}_{ab}^{X;p_1 p_2}$ on the right-hand side is one of the You seeds in Appendix A; hence every term is an explicit combination of Gamma functions and ${}_4\tilde{F}_3(-1)$. Equation (75), analytically continued from the regulated half-plane, is also a useful numerical representation of the Hilbert transforms below. At $u = v = 0$ it reduces to the finite four-seed combination in Eq. (44) before the SK sum.

Let $\epsilon_{23}, \epsilon_{41} \in \{0, 1\}$ indicate whether the algebraic or sign term in Eq. (73) is selected. Define

$$\mathbb{P}_{\epsilon_{23}\epsilon_{41}} F \equiv \prod_{e \in \{23, 41\}; \epsilon_e=1} \left[\frac{1}{i\pi} \text{PV} \int_{-\infty}^{+\infty} \frac{d\Omega_e}{\Omega_e} \right] F \left(\frac{2\epsilon_{41}\Omega_{41}}{k_{12}}, -\frac{2\epsilon_{23}\Omega_{23}}{k_{12}}, \frac{2\epsilon_{23}\Omega_{23}}{k_{34}}, -\frac{2\epsilon_{41}\Omega_{41}}{k_{34}} \right), \quad (76)$$

where an absent Ω_e is set to zero. The four arguments of F are respectively $(u_L, v_L; u_R, v_R)$. They follow directly from $e^{i\Omega_{23}(\tau_2-\tau_3)}$ and $e^{i\Omega_{41}(\tau_4-\tau_1)}$.

For row ℓ , decompose the two branch-resolved hard blocks as

$$\begin{aligned} \mathbf{Q}_{+;ab}^{X_L}(\hat{\mathbf{k}}_1; u_L, v_L) &= L_{0;ab}^\ell \sigma^0 + L_{1;ab}^\ell \hat{\mathbf{k}}_1 \cdot \sigma, \\ \mathbf{Q}_{+;cd}^{X_R}(\hat{\mathbf{k}}_3; u_R, v_R) &= R_{0;cd}^\ell \sigma^0 + R_{1;cd}^\ell \hat{\mathbf{k}}_3 \cdot \sigma. \end{aligned} \quad (77)$$

As in Eq. (48), the sign appropriate to a dotted block has already been absorbed into L_1 or R_1 . For $m, n = 0, 1$, define the exact hard functional

$$\mathcal{H}_\ell^{mn} \equiv \sum_{a,b,c,d=\pm} abcd \sum_{\epsilon_{23}, \epsilon_{41}} C_{bc}^{(\epsilon_{23})} C_{da}^{(\epsilon_{41})}, \mathbb{P}_{\epsilon_{23}\epsilon_{41}} \left[L_{m;ab}^\ell R_{n;cd}^\ell \right]. \quad (78)$$

For a type-2/type-3 soft edge, the corresponding ϵ sum in Eq. (78) is replaced by the single value $\epsilon = 0$ with $C^{(0)} = 1$. For a type-1 edge both values are retained. Thus $\mathcal{H}_\ell^{mn}$ contains a finite piece and at most one PV integral for AN, NA , and at most two PV integrals for NN . Equations (75), (76), and (78) are the corrected branch-resolved form of the antisymmetric master formula. To write it uniformly, set

$$\chi_e^\ell = \begin{cases} \Xi_e, & \text{if edge } e \text{ is type-1 in row } \ell, \\ 1, & \text{if edge } e \text{ is type-2/type-3 in row } \ell. \end{cases} \quad (79)$$

Here Σ_e is the raised soft spinor matrix: it is σ^0 on a type-2/type-3 edge, and $\zeta_e \hat{\mathbf{p}}_e \cdot \sigma$ on a type-1 edge, with $\mathbf{p}_{23} = \mathbf{K}$ and $\mathbf{p}_{41} = \mathbf{q}$. The bracket $[\cdots]_{\text{spectral}}$ means that the vector or tensor bubble moment in Eq. (69) and the Hilbert operators in Eq. (76) are applied before the branch sum. Then

$$\mathcal{W}_{+,\ell}^{(A)} = \sum_{a,b,c,d=\pm} abcd \left[\chi_{23;bc}^\ell \chi_{41;da}^\ell \text{tr}_2 (\mathbf{Q}_{+;ab}^{X_L} \Sigma_{23} \mathbf{Q}_{+;cd}^{X_R} \Sigma_{41}) \right]_{\text{spectral}}. \quad (80)$$

4.4 Spinor contractions for the twelve nonzero rows

Write

$$\begin{aligned} c_1 &= \hat{\mathbf{k}}_1 \cdot \hat{\mathbf{k}}_s = \cos \theta_1, & c_3 &= \hat{\mathbf{k}}_3 \cdot \hat{\mathbf{k}}_s = \cos \theta_3, \\ p_{13} &= \hat{\mathbf{k}}_s \cdot (\hat{\mathbf{k}}_1 \times \hat{\mathbf{k}}_3) = \sin \theta_1 \sin \theta_3 \sin \phi_3. \end{aligned} \quad (81)$$

For $\ell \in \mathcal{G}_{AN}$, the type-1 line is (41) and the Pauli matrix lies to the right of the right hard block. Its contraction is

$$\mathcal{W}_\ell^{AN} = 2\zeta_{41}^\ell \mathcal{V}_q \left(\mathcal{H}_\ell^{01} c_3 + \mathcal{H}_\ell^{10} c_1 + i\mathcal{H}_\ell^{11} p_{13} \right), \quad \ell = 2, 8, 9, 15. \quad (82)$$

For $\ell \in \mathcal{G}_{NA}$, the type-1 matrix is between the two hard blocks, giving the opposite sign for the three-Pauli trace,

$$\mathcal{W}_\ell^{NA} = 2\zeta_{23}^\ell \mathcal{V}_K \left(\mathcal{H}_\ell^{01} c_3 + \mathcal{H}_\ell^{10} c_1 - i\mathcal{H}_\ell^{11} p_{13} \right), \quad \ell = 3, 5, 12, 14. \quad (83)$$

Finally, contracting the symmetric tensor in Eq. (69) with $\text{tr}(\mathbf{Q}_L \sigma^i \mathbf{Q}_R \sigma^j)$ gives

$$\mathcal{W}_\ell^{NN} = 2\zeta_{23}^\ell \zeta_{41}^\ell \left\{ \mathcal{J}_0 \mathcal{H}_\ell^{00} + \mathcal{H}_\ell^{11} \left[-\mathcal{J}_L c_{13} + 2(\mathcal{J}_L - \mathcal{J}_T) c_1 c_3 \right] \right\}, \quad \ell = 4, 6, 11, 13. \quad (84)$$

All bubble functions in Eqs. (82)–(84) are evaluated at $x = -i\tilde{\nu}$. The signs read directly from the table:

$$\begin{aligned} (\zeta_{41}^2, \zeta_{41}^8, \zeta_{41}^9, \zeta_{41}^{15}) &= (-1, -1, +1, +1), \\ (\zeta_{23}^3, \zeta_{23}^5, \zeta_{23}^{12}, \zeta_{23}^{14}) &= (+1, -1, +1, -1), \\ (\zeta_{23} \zeta_{41})_{4,6,11,13} &= (-1, +1, +1, -1). \end{aligned} \quad (85)$$

Equations (82)–(85), together with the hard-pair column of the table, are the individual result for each of the twelve nonzero rows. No spinor index or loop momentum remains uncontracted.

For clarity, the twelve answers can be displayed without any class notation. Define

$$\begin{aligned} F_\pm^\ell &= \mathcal{H}_\ell^{01} c_3 + \mathcal{H}_\ell^{10} c_1 \pm i\mathcal{H}_\ell^{11} p_{13}, \\ G^\ell &= \mathcal{J}_0 \mathcal{H}_\ell^{00} + \mathcal{H}_\ell^{11} \left[-\mathcal{J}_L c_{13} + 2(\mathcal{J}_L - \mathcal{J}_T) c_1 c_3 \right]. \end{aligned} \quad (86)$$

The superscript ℓ fixes the hard pair listed in the sixteen-row table, so that, for example, $\mathcal{H}_2^{mn} = \mathcal{H}^{mn}[\widehat{D}, D]$ and $\mathcal{H}_{13}^{mn} = \mathcal{H}^{mn}[\check{D}, \widehat{D}]$. The nonzero contractions are

$$\begin{aligned} \mathcal{W}_2^{AN} &= -2\mathcal{V}_q F_+^2, & \mathcal{W}_8^{AN} &= -2\mathcal{V}_q F_+^8, & \mathcal{W}_9^{AN} &= +2\mathcal{V}_q F_+^9, & \mathcal{W}_{15}^{AN} &= +2\mathcal{V}_q F_+^{15}, \\ \mathcal{W}_3^{NA} &= +2\mathcal{V}_K F_-^3, & \mathcal{W}_5^{NA} &= -2\mathcal{V}_K F_-^5, & \mathcal{W}_{12}^{NA} &= +2\mathcal{V}_K F_-^{12}, & \mathcal{W}_{14}^{NA} &= -2\mathcal{V}_K F_-^{14}, \\ \mathcal{W}_4^{NN} &= -2G^4, & \mathcal{W}_6^{NN} &= +2G^6, & \mathcal{W}_{11}^{NN} &= +2G^{11}, & \mathcal{W}_{13}^{NN} &= -2G^{13}. \end{aligned} \quad (87)$$

Together with $\mathcal{W}_{1,7,10,16}^{(A)} = 0$, Eq. (87) is an explicit sixteen-row audit of the antisymmetric sector.

4.5 Nonlocal signal and angular coefficients

One symmetric type-2/type-3 line contributes $\kappa_+ = i\rho_+$ and one antisymmetric type-1 line contributes $-\rho_+$. Two type-1 lines contribute ρ_+^2 . It follows that

$$C_{(A)}^{\text{Box}} = \mathcal{P}_A(\tilde{\nu}) \left[-i \sum_{\ell \in \mathcal{G}_{AN}} \mathcal{W}_\ell^{AN} - i \sum_{\ell \in \mathcal{G}_{NA}} \mathcal{W}_\ell^{NA} + \sum_{\ell \in \mathcal{G}_{NN}} \mathcal{W}_\ell^{NN} \right], \quad \mathcal{P}_A(\tilde{\nu}) = \frac{2^{12i\tilde{\nu}} \rho_+^2}{(4\pi)^{3/2}}. \quad (88)$$

This already has a form directly suitable for numerical evaluation. To display the angle dependence as in the gauge-boson box, write

$$C_{(A)}^{\text{Box}} = A_0^{(A)} + A_1^{(A)} c_1 + A_3^{(A)} c_3 + A_{13}^{(A)} c_{13} + A_L^{(A)} c_1 c_3 + A_P^{(A)} p_{13}. \quad (89)$$

The six mass- and ratio-dependent coefficients are

$$A_1^{(A)} = -2i\mathcal{P}_A \left[\sum_{\ell \in \mathcal{G}_{AN}} \zeta_{41}^\ell \mathcal{V}_q \mathcal{H}_\ell^{10} + \sum_{\ell \in \mathcal{G}_{NA}} \zeta_{23}^\ell \mathcal{V}_K \mathcal{H}_\ell^{10} \right], \quad (90a)$$

$$A_3^{(A)} = -2i\mathcal{P}_A \left[\sum_{\ell \in \mathcal{G}_{AN}} \zeta_{41}^\ell \mathcal{V}_q \mathcal{H}_\ell^{01} + \sum_{\ell \in \mathcal{G}_{NA}} \zeta_{23}^\ell \mathcal{V}_K \mathcal{H}_\ell^{01} \right], \quad (90b)$$

$$A_0^{(A)} = 2\mathcal{P}_A \sum_{\ell \in \mathcal{G}_{NN}} \zeta_{23}^\ell \zeta_{41}^\ell \mathcal{J}_0 \mathcal{H}_\ell^{00}, \quad (90c)$$

$$A_{13}^{(A)} = -2\mathcal{P}_A \sum_{\ell \in \mathcal{G}_{NN}} \zeta_{23}^\ell \zeta_{41}^\ell \mathcal{J}_L \mathcal{H}_\ell^{11}, \quad (90d)$$

$$A_L^{(A)} = 4\mathcal{P}_A \sum_{\ell \in \mathcal{G}_{NN}} \zeta_{23}^\ell \zeta_{41}^\ell (\mathcal{J}_L - \mathcal{J}_T) \mathcal{H}_\ell^{11}, \quad (90e)$$

$$A_P^{(A)} = 2\mathcal{P}_A \left[\sum_{\ell \in \mathcal{G}_{AN}} \zeta_{41}^\ell \mathcal{V}_q \mathcal{H}_\ell^{11} - \sum_{\ell \in \mathcal{G}_{NA}} \zeta_{23}^\ell \mathcal{V}_K \mathcal{H}_\ell^{11} \right]. \quad (90f)$$

Here a sum labelled by a class runs over the four rows in Eq. (71). The coefficients depend on $\tilde{\nu}$ through ρ_+ , the Gamma-function bubble seeds, and the Gamma $_{-4}\tilde{F}_3$ series in Eq. (75). They also depend on the hard-scale ratio $r = k_{12}/k_{34}$ through the frequency arguments in Eq. (76). Thus their mass and kinematic dependence is explicit.

Charge conjugation pairs the rows as (2, 15), (8, 9), (3, 14), (5, 12), (4, 13), and (6, 11). Applying Eq. (47) inside Eq. (78), followed where necessary by the harmless PV change of variables $\Omega_e \rightarrow -\Omega_e$, shows that their parity-odd pieces cancel pairwise, so

$$A_P^{(A)} = 0. \quad (91)$$

The exact antisymmetric answer for one fixed cyclic ordering is therefore the five-term polynomial in Eq. (89) with its last term omitted. This is the most direct analogue of the gauge-boson angular decomposition.

The complete answer for a fixed cyclic ordering is

$$\langle \delta h^4 \rangle'_{\text{Box}} = -N_c g_t^4 \frac{H^4 k_s^3}{16 k_1^3 k_2^3 k_3^3 k_4^3} \left(\frac{k_s^2}{4 k_{12} k_{34}} \right)^{2i\tilde{\nu}} C^{\text{Box}}(\tilde{\nu}, r, \theta_1, \theta_3, \phi_3) + \text{c.c.}, \quad (92)$$

$$\begin{aligned} C^{\text{Box}} &= C_{(S)}^{\text{Box}} + C_{(A)}^{\text{Box}} \\ &= A_0 + A_1 c_1 + A_3 c_3 + A_{13} c_{13} + A_L c_1 c_3, \end{aligned} \quad (93)$$

where

$$\begin{aligned} A_0 &= A_0^{(S)} + A_0^{(A)}, & A_{13} &= A_{13}^{(S)} + A_{13}^{(A)}, \\ A_1 &= A_1^{(A)}, & A_3 &= A_3^{(A)}, & A_L &= A_L^{(A)}. \end{aligned} \quad (94)$$

The complex power is identical to the symmetric and gauge-boson boxes. The only new objects are one- and two-dimensional Hilbert transforms of known fermion seeds. A finite Gamma-function expression for these transforms is not available in Ref. [2]; that paper explicitly identifies the antisymmetric time ordering as the obstruction to direct cutting. Equations (75) and (76) isolate this obstruction without leaving any four-time integral.

For later reference, the symmetric angular result is

$$C_{(S)}^{\text{Box}} = A_0^{(S)} + A_{13}^{(S)} c_{13}. \quad (95)$$

Combining it with Eq. (91), the angular dependence of one cyclic box is

$$C_{\text{cyclic}}^{\text{Box}} = A_0 + A_{13} c_{13} + A_1 c_1 + A_3 c_3 + A_L c_1 c_3. \quad (96)$$

There is no $\cos 2\phi_3$ term.

The observable correlator of identical Higgs fluctuations also includes crossed cyclic orderings. At leading order, exchanging the two legs in the left pair sends $\hat{\mathbf{k}}_1 \rightarrow -\hat{\mathbf{k}}_1$, and the analogous exchange on the right sends $\hat{\mathbf{k}}_3 \rightarrow -\hat{\mathbf{k}}_3$. Averaging independently over these exchanges projects Eq. (96) onto

$$C_{\text{Bose}}^{\text{Box}} = A_0(\tilde{\nu}, r) + \mathcal{O}((k_s/k_i)). \quad (97)$$

Hence the leading, fully Bose-symmetrized top-box clock signal is isotropic. The angular terms in Eq. (96) belong to an ordered box and must not be confused with the final identical-scalar angular dependence.

The scale dependence in the gauge-boson convention is

$$k_s^3 \left(\frac{k_s^2}{4k_{12}k_{34}} \right)^{2i\tilde{\nu}} + \text{c.c.} \sim k_s^3 \cos \left[2\tilde{\nu} \log \left(\frac{k_s^2}{4k_{12}k_{34}} \right) + \varphi \right]. \quad (98)$$

5 Independent checks

We close with checks that do not reuse the factorized derivation.

First, substituting Eq. (4) directly into the Dirac action reproduces both the mass term and the Yukawa vertex in Eq. (6). The transformation is unitary, so its Grassmann Jacobian is one. The two Majorana determinants reproduce one Dirac determinant, confirming the N_c , rather than $2N_c$, normalization.

Second, the lower-index type-2/type-3 helicity sum is the invariant tensor ϵ . Raising one index gives the identity, proving the rotation independence and the SK equality used in the cut. This also shows directly that no loop-momentum angular integral was silently dropped in Eq. (25).

Third, enumerating the two vertex choices as four-bit words gives exactly sixteen entries. The edge rule in Eq. (16) gives four entries in each of the AA , AN , NA , and NN classes, so no chirality flow is duplicated or omitted.

Fourth, the conformal loop formula gives $k_s^{3+4if\tilde{\nu}}$. The two hard subgraphs give

$$\left(\frac{k_{12}}{2}\right)^{-2if\tilde{\nu}} \left(\frac{k_{34}}{2}\right)^{-2if\tilde{\nu}}, \quad (99)$$

so Eq. (92) has the stated nonanalytic ratio and overall momentum dimension k^{-9} .

Fifth, Eqs. (61) and (72) were checked distributionally: $d \operatorname{sgn}(t)/dt = 2\delta(t)$ and the principal-value transform gives the same derivative. This independently fixes the signs of all four SK kernels.

Sixth, multiplying the four pairs of 2×2 matrices in the $\mathbf{k} \parallel \hat{\mathbf{z}}$ frame and then applying the $SU(2)$ rotations reproduces all four traces in Eq. (43). This independently leaves only 1 and c_{13} in the symmetric sector. For the antisymmetric sector, direct Pauli-matrix multiplication gives Eqs. (82), (83), and (84), including the opposite signs of the two triple-product terms. Pairing every word with its charge conjugate then eliminates that triple product. The tensor identity $2\mathcal{J}_T + \mathcal{J}_L = \mathcal{J}_0$ follows by tracing Eq. (69); it also follows independently from $\hat{\mathbf{q}} \cdot \hat{\mathbf{K}} = (q^2 + K^2 - k_s^2)/(2qK)$.

Seventh, multiplying the two late-time soft factors directly gives

$$(4q^2\tau_2\tau_3)^{if\tilde{\nu}}(4K^2\tau_4\tau_1)^{if\tilde{\nu}} = 2^{4if\tilde{\nu}} q^{2if\tilde{\nu}} K^{2if\tilde{\nu}} \prod_{j=1}^4 (-\tau_j)^{if\tilde{\nu}}. \quad (100)$$

This verifies independently the separation into the bubble subgraph and the four extra time powers in Eqs. (23) and (70).

Finally,

$$|\kappa_+(\tilde{\nu})| = \left| \frac{\Gamma(-2i\tilde{\nu})}{\Gamma(-i\tilde{\nu})} \right|^2 = \frac{1}{4 \cosh(\pi\tilde{\nu})}, \quad (101)$$

so the two cut soft lines contain the expected large-mass suppression $|\kappa_+|^2 \sim e^{-2\pi\tilde{\nu}}/4$. Moreover $\kappa_- = \kappa_+^*$, $\mathcal{Q}_- = \mathcal{Q}_+^*$, and $f_0(i\tilde{\nu}, i\tilde{\nu}) = f_0(-i\tilde{\nu}, -i\tilde{\nu})^*$, which proves that the $f = -$ branch is the complex conjugate displayed in Eq. (92).

As a final seed check, setting $\tilde{\lambda} = 0$ and $\tilde{M}_R = \tilde{\nu}$ in Eqs. (2.52), (2.54), and (2.58) of Ref. [2] gives Eqs. (49)–(51) term by term. The reversal and complex-conjugation identities in Eq. (47) then reconstruct the $\bar{D}$ and $\check{D}$ blocks, so the four symmetric diagrams require no seed beyond the three functions $\mathfrak{I}$, $\mathfrak{J}$, and $\mathfrak{K}$. Expanding the energy deformation as in Eq. (75) proves the same statement term by term for all twelve nonzero antisymmetric diagrams.

A Closed form of the hard fermion seed

For completeness, this appendix collects the type-1, type-2, and type-3 hard seeds at zero chemical potential, using the integral formulae of Ref. [2]. They make the four surviving terms in Eq. (56) explicit in Gamma and generalized hypergeometric functions. The antisymmetric contribution remains in the exact principal-value form of Eq. (80).

For $\mathbf{k} \parallel \hat{\mathbf{z}}$, write

$$\mathcal{I}_{++}^{p_1 p_2} = \begin{pmatrix} A_1^{p_1 p_2} - B_2^{p_1 p_2} & 0 \\ 0 & A_2^{p_1 p_2} - B_1^{p_1 p_2} \end{pmatrix}, \quad (102)$$

$$\mathcal{I}_{--}^{p_1 p_2} = \left(\mathcal{I}_{++}^{p_2^* p_1^*} \right)^\dagger, \quad (103)$$

$$\mathcal{I}_{-+}^{p_1 p_2} = \begin{pmatrix} U_1^{p_1 p_2} & 0 \\ 0 & U_2^{p_1 p_2} \end{pmatrix}, \quad \mathcal{I}_{+-}^{p_1 p_2} = \begin{pmatrix} -V_2^{p_1 p_2} & 0 \\ 0 & -V_1^{p_1 p_2} \end{pmatrix}. \quad (104)$$

The entries are

$$A_1^{p_1 p_2} = \tilde{\nu}^2 \mathfrak{I}^{p_1 p_2}(1 + i\tilde{\nu}, 1 - i\tilde{\nu}, i\tilde{\nu}), \quad (105)$$

$$A_2^{p_1 p_2} = \mathfrak{I}^{p_1 p_2}(i\tilde{\nu}, -i\tilde{\nu}, 1 + i\tilde{\nu}), \quad (106)$$

$$B_1^{p_1 p_2} = \mathfrak{I}^{p_2 p_1}(i\tilde{\nu}, -i\tilde{\nu}, 1 + i\tilde{\nu}), \quad (107)$$

$$B_2^{p_1 p_2} = \tilde{\nu}^2 \mathfrak{I}^{p_2 p_1}(1 + i\tilde{\nu}, 1 - i\tilde{\nu}, i\tilde{\nu}), \quad (108)$$

$$U_1^{p_1 p_2} = \tilde{\nu}^2 \mathfrak{K}^{p_1 p_2}(2, 2), \quad U_2^{p_1 p_2} = \mathfrak{K}^{p_1 p_2}(1, 1), \quad (109)$$

$$V_1^{p_1 p_2} = e^{-i\pi(p_1 - p_2)} \mathfrak{K}^{p_1 p_2}(1, 1), \quad V_2^{p_1 p_2} = \tilde{\nu}^2 e^{-i\pi(p_1 - p_2)} \mathfrak{K}^{p_1 p_2}(2, 2). \quad (110)$$

The factorized seed is

$$\mathfrak{K}^{p_1 p_2}(x, y) = e^{\frac{i\pi}{2}(p_1 - p_2)} \frac{\Gamma(1 + p_1 - i\tilde{\nu})\Gamma(1 + p_1 + i\tilde{\nu})\Gamma(1 + p_2 - i\tilde{\nu})\Gamma(1 + p_2 + i\tilde{\nu})}{2^{2+p_1+p_2}\Gamma(p_1 + x)\Gamma(p_2 + y)}. \quad (111)$$

The nested seed is

$$\begin{aligned} \mathfrak{I}^{p_1 p_2}(x, y, z) &= i 2^{-2-p_1-p_2} e^{-\frac{i\pi}{2}(p_1+p_2)} \pi \operatorname{csch}(2\pi\tilde{\nu}) \Gamma(2 + p_1 + p_2) \\ &\times \left[\frac{e^{-\pi\tilde{\nu}}}{\Gamma(x)} \Gamma(1 + p_1 - i\tilde{\nu}) \Gamma(2 + p_1 + p_2 - 2i\tilde{\nu}) \right. \\ &\times {}_4\tilde{F}_3 \left(\begin{matrix} 2 + p_1 + p_2, & 1 + p_1 - i\tilde{\nu}, & y, & 2 + p_1 + p_2 - 2i\tilde{\nu} \\ 2 + p_1 - i\tilde{\nu}, & 3 + p_1 + p_2 - z, & 1 - 2i\tilde{\nu} \end{matrix}; -1 \right) - (\tilde{\nu} \rightarrow -\tilde{\nu}) \Big]. \quad (112) \end{aligned}$$

Here ${}_4\tilde{F}_3$ denotes the regularized generalized hypergeometric function. Equations (104)–(112), inserted into the $rs = N\bar{N}$ case of Eq. (33) with $p_+ = -1 + i\tilde{\nu}$, determine $Q_{+,0}$ and $Q_{+,1}$ without any remaining time integral. The reversed $rs = \bar{N}N$ block follows from Eq. (17).

For the type-2 hard propagator, the corresponding lower-index matrices are

$$\hat{\mathcal{I}}_{++}^{p_1 p_2} = \begin{pmatrix} 0 & S_1^{p_1 p_2} + S_2^{p_2 p_1} \\ -(S_1^{p_2 p_1} + S_2^{p_1 p_2}) & 0 \end{pmatrix}, \quad (113)$$

$$\hat{\mathcal{I}}_{--}^{p_1 p_2} = \begin{pmatrix} 0 & S_3^{p_1 p_2} + S_4^{p_2 p_1} \\ -(S_3^{p_2 p_1} + S_4^{p_1 p_2}) & 0 \end{pmatrix}, \quad (114)$$

$$\hat{\mathcal{I}}_{+-}^{p_1 p_2} = \begin{pmatrix} 0 & W_1^{p_1 p_2} \\ -W_2^{p_1 p_2} & 0 \end{pmatrix}, \quad \hat{\mathcal{I}}_{-+}^{p_1 p_2} = \begin{pmatrix} 0 & W_2^{p_2 p_1} \\ -W_1^{p_2 p_1} & 0 \end{pmatrix}. \quad (115)$$

At zero chemical potential their entries reduce to

$$S_1^{p_1 p_2} = \tilde{\nu} \mathfrak{J}^{p_1 p_2}(1 + i\tilde{\nu}, 1 - i\tilde{\nu}, 1 + i\tilde{\nu}), \quad (116)$$

$$S_2^{p_1 p_2} = \tilde{\nu} \mathfrak{J}^{p_1 p_2}(i\tilde{\nu}, -i\tilde{\nu}, i\tilde{\nu}), \quad (117)$$

$$S_3^{p_1 p_2} = \tilde{\nu} \mathfrak{J}^{p_2 p_1}(i\tilde{\nu}, -i\tilde{\nu}, i\tilde{\nu}), \quad (118)$$

$$S_4^{p_1 p_2} = \tilde{\nu} \mathfrak{J}^{p_2 p_1}(1 + i\tilde{\nu}, 1 - i\tilde{\nu}, 1 + i\tilde{\nu}), \quad (119)$$

$$W_1^{p_1 p_2} = \tilde{\nu} \mathfrak{R}^{p_1 p_2}(2, 1), \quad W_2^{p_1 p_2} = \tilde{\nu} \mathfrak{R}^{p_1 p_2}(1, 2). \quad (120)$$

The second nested function is

$$\begin{aligned} \mathfrak{J}^{p_1 p_2}(x, y, z) &= i 2^{-2-p_1-p_2} e^{\frac{i\pi}{2}(p_1+p_2)} \pi \operatorname{csch}(2\pi\tilde{\nu}) \Gamma(2+p_1+p_2) \\ &\times \left[\frac{e^{\pi\tilde{\nu}}}{\Gamma(x)} \Gamma(1+p_1-i\tilde{\nu}) \Gamma(2+p_1+p_2-2i\tilde{\nu}) \right. \\ &\times {}_4\tilde{F}_3 \left(\begin{matrix} 2+p_1+p_2, 1+p_1-i\tilde{\nu}, y, 2+p_1+p_2-2i\tilde{\nu} \\ 2+p_1-i\tilde{\nu}, 3+p_1+p_2-z, 1-2i\tilde{\nu} \end{matrix}; -1 \right) - (\tilde{\nu} \rightarrow -\tilde{\nu}) \Big]. \end{aligned} \quad (121)$$

The type-3 seed follows without another integration,

$$\check{\mathcal{I}}_{ab}^{p_1 p_2} = - \left(\hat{\mathcal{I}}_{-a, -b}^{p_1^* p_2^*} \right)^*. \quad (122)$$

Raising the appropriate second index and using Eqs. (115)–(122) in the corresponding $rs = NN$ and $rs = \bar{N}\bar{N}$ cases of Eq. (33) supplies the type-2 and type-3 hard blocks in Eq. (56). Thus all four entries of $\mathcal{G}_{AA}$ are explicit.

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